



CUSTOMER CHURN PREDICTION USING MACHINE LEARNING

Artificial Intelligence Internship – Task 1 Report

ML Classification Project

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1. Introduction

Customer retention plays an important role in the growth and success of a business. In the telecommunications industry, customer churn is a major challenge because losing customers directly affects revenue and business performance. Identifying customers who are likely to leave the service allows companies to take preventive actions and improve customer satisfaction.

This project focuses on developing a Machine Learning-based Customer Churn Prediction system that predicts whether a telecom customer will continue using the service or discontinue it.

The project uses supervised machine learning classification techniques to analyze customer information such as demographic details, subscribed services, account

information, and billing details. By identifying patterns in customer behavior, the model helps predict potential churn customers.

This project demonstrates the practical application of Artificial Intelligence concepts including data preprocessing, exploratory data analysis, model training, evaluation, and model saving.

2. Objective

The main objective of this project is to develop a machine learning classification model that can accurately predict customer churn.

The objectives of this project are:

- To understand and analyze customer-related data.
 - To perform data cleaning and preprocessing.
 - To explore customer behavior using visualization techniques.
 - To transform raw data into a format suitable for machine learning.
 - To train classification models using customer data.
 - To compare different machine learning algorithms.
 - To evaluate model performance using standard evaluation metrics.
 - To select the best-performing model for churn prediction.
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3. Problem Statement

Telecommunication companies generate large amounts of customer data. However, manually identifying customers who may discontinue their services is difficult and time-consuming.

The objective of this project is to build a predictive model that can classify customers **into:**

Churn Customers: Customers who are likely to leave the telecom service.

Non-Churn Customers: Customers who are likely to continue using the service.

The developed model can help companies identify possible churn customers and take suitable retention measures.

4. Dataset Description

The dataset used in this project is the **Telco Customer Churn Dataset**.

The dataset contains customer information related to:

- Customer demographics
- Account information
- Internet and phone services
- Contract details
- Payment methods
- Monthly charges
- Total charges
- Customer churn status

Dataset Details:

- Total Number of Records: 7043
- Total Number of Features: 21
- Target Variable: Churn

The target variable contains two categories:

- Yes → Customer has churned
 - No → Customer has not churned
-

5. Technologies Used

Programming Language

Python 3.x

Python was used because of its wide range of machine learning libraries and easy implementation.

Libraries Used:

Pandas

Used for:

- Data loading
- Data manipulation
- Data analysis

NumPy

Used for:

- Numerical operations
- Array processing

Matplotlib and Seaborn

Used for:

- Data visualization
- Exploratory Data Analysis
- Creating graphs and charts

Scikit-Learn

Used for:

- Data preprocessing
- Model building
- Model evaluation

Joblib

Used for:

- Saving the trained machine learning model

Jupyter Notebook

Used as the development environment for writing and executing Python code.

6. Project Workflow

The complete workflow followed in this project is:

Step 1: Data Loading

The Telco Customer Churn dataset was imported into the Python environment using Pandas.

Step 2: Data Cleaning

The dataset was checked for:

- Missing values

- Incorrect data types
- Duplicate records

Step 3: Exploratory Data Analysis

Various visualizations were created to understand customer patterns.

Step 4: Feature Engineering

Categorical data was converted into numerical format.

Step 5: Data Splitting

The dataset was divided into training and testing data.

Step 6: Model Training

Classification algorithms were trained using the training dataset.

Step 7: Model Evaluation

Performance was measured using different evaluation metrics.

Step 8: Model Saving

The final trained model was saved for future prediction.

7. Data Preprocessing

Data preprocessing is an important step in machine learning because raw data cannot be directly used by algorithms.

The following preprocessing techniques were performed:

Handling Missing Values

The dataset was checked for missing values.

The TotalCharges column contained blank values, which were handled before model training.

Data Type Conversion

The TotalCharges feature was converted from object format into numerical format.

Feature Encoding

Machine learning algorithms require numerical inputs. Therefore, categorical features were converted using Label Encoding.

Examples:

- Gender
- Contract Type
- Internet Service
- Payment Method
- Churn Status

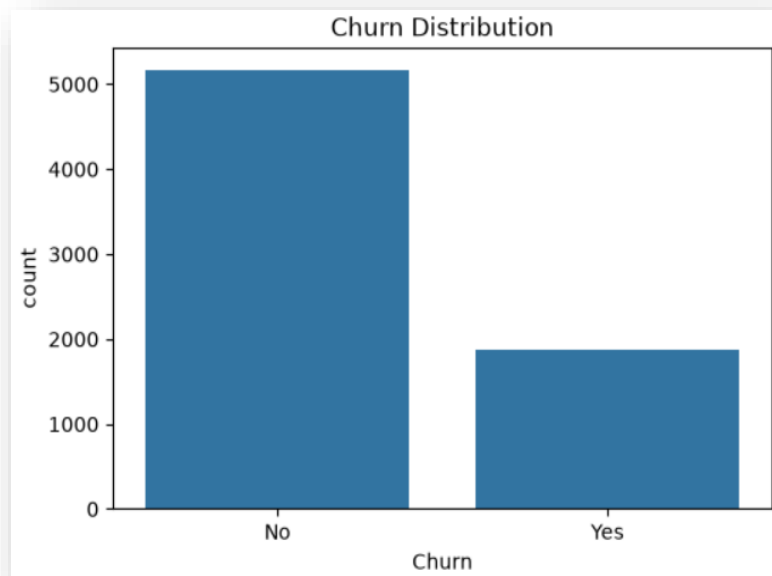
8. Exploratory Data Analysis

Exploratory Data Analysis was performed to understand the dataset and identify important patterns.

The following analysis was performed:

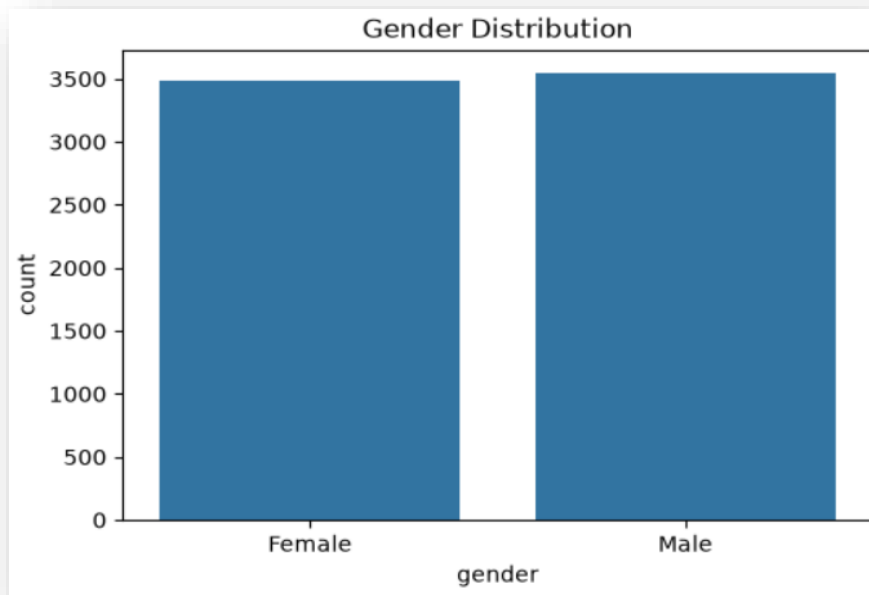
Churn Distribution

The distribution of churn and non-churn customers was analyzed.



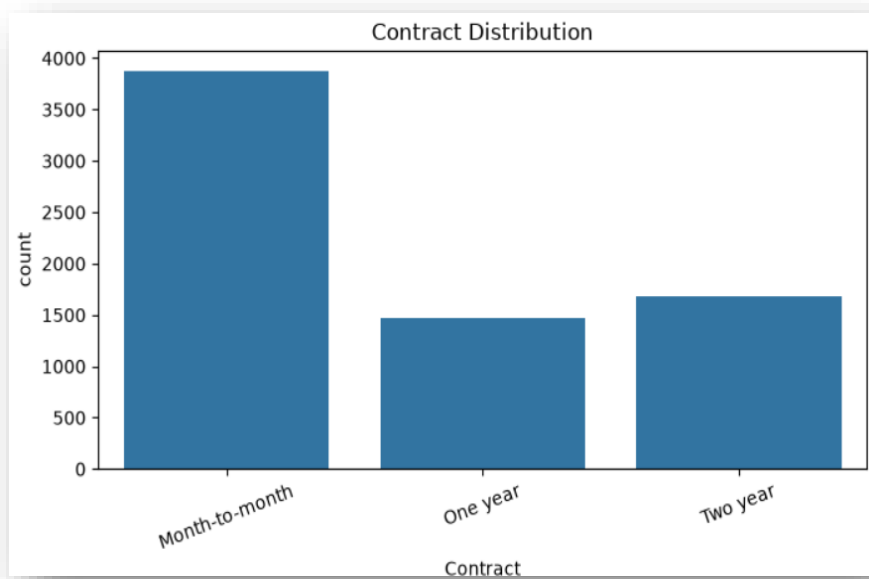
Gender Analysis

Customer distribution based on gender was visualized.



Contract Analysis

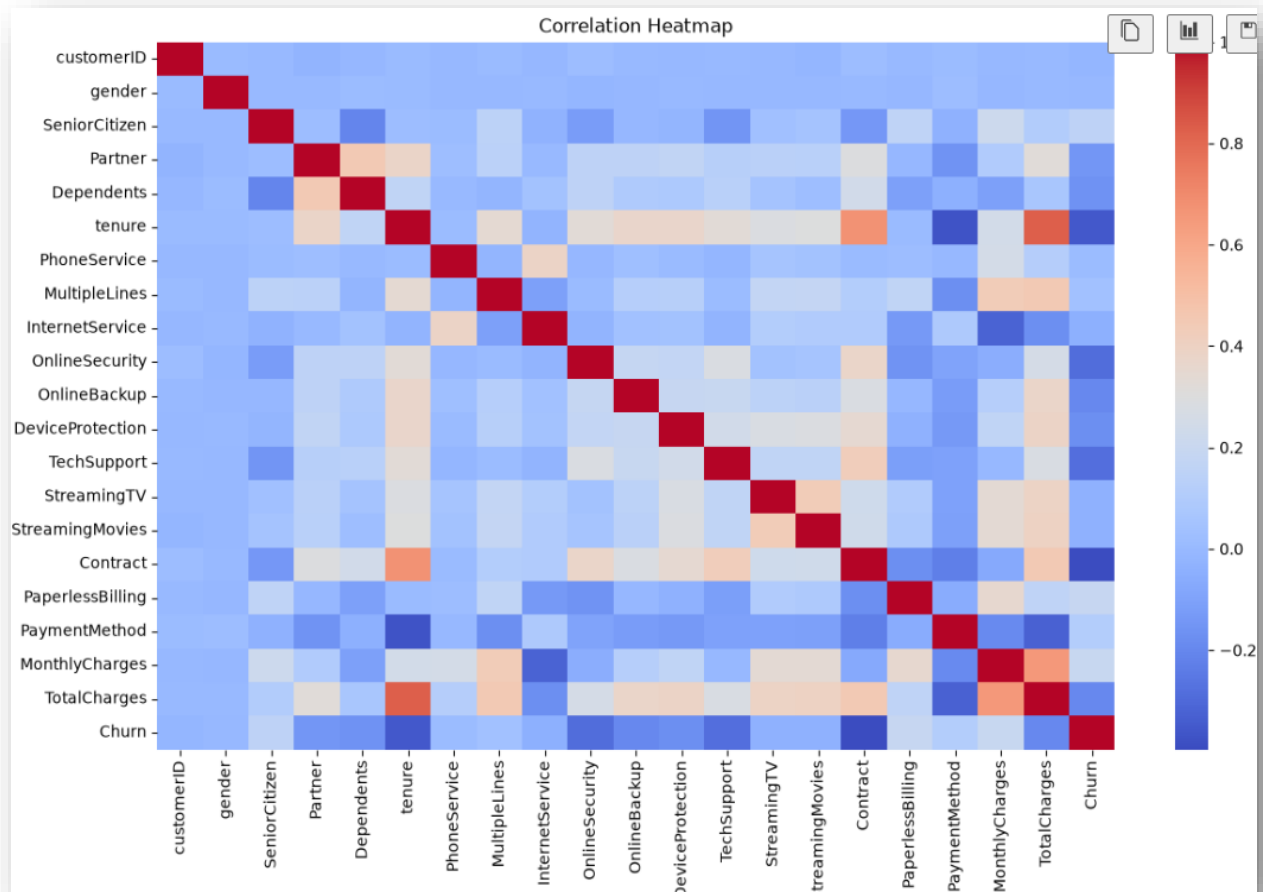
The relationship between contract type and churn probability was analyzed.



Correlation Analysis

Feature relationships were analyzed using correlation visualization.

EDA helped in understanding customer characteristics that influence churn behavior.



9. Machine Learning Model Development

Two classification algorithms were implemented and compared.

1. Logistic Regression

Logistic Regression is a supervised learning algorithm used for binary classification problems.

In this project, it was used to predict whether a customer belongs to the churn or non-churn category.

Advantages:

- Simple implementation
 - Easy interpretation
 - Suitable for classification problems
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2. Random Forest Classifier

Random Forest is an ensemble machine learning algorithm that combines multiple decision trees to improve prediction performance.

Advantages:

- Handles complex relationships
 - Reduces overfitting
 - Provides feature importance analysis
 - Works effectively with customer datasets
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10. Data Splitting

The dataset was divided into training and testing datasets using an 80:20 ratio.

Training Dataset

Used for training the machine learning models.

Number of Records:

5625

Testing Dataset

Used for evaluating the performance of trained models.

Number of Records:

1407

11. Model Evaluation

The models were evaluated using the following performance metrics:

Accuracy Score

Accuracy represents the percentage of correct predictions made by the model.

Precision Score

Precision measures how many predicted churn customers were actually churn customers.

Recall Score

Recall measures how many actual churn customers were successfully identified.

F1 Score

F1 score provides a balance between precision and recall.

ROC-AUC Score

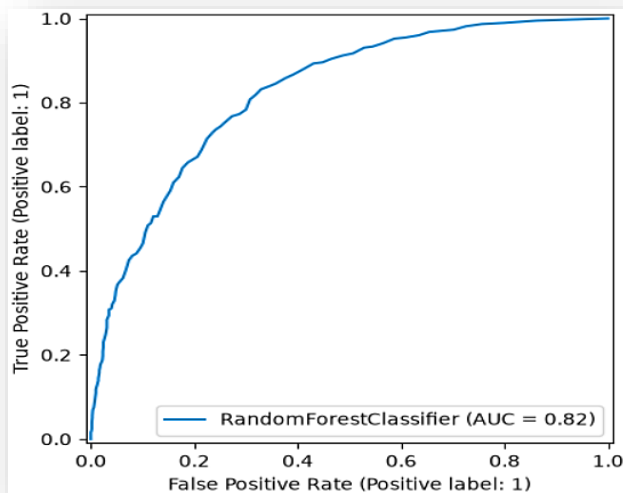
ROC-AUC evaluates the ability of the model to differentiate between churn and non-churn customers.

LOGISTIC REGRESSION RESULTS

Accuracy : 0.7874911158493249
Precision: 0.6262626262626263
Recall : 0.49732620320855614
F1 Score : 0.5543964232488823
ROC AUC : 0.6949360928917901

RANDOM FOREST RESULTS

Accuracy : 0.7846481876332623
Precision: 0.6319702602230484
Recall : 0.45454545454545453
F1 Score : 0.5287713841368584
ROC AUC : 0.6793540438264544



12. Model Comparison and Selection

Both Logistic Regression and Random Forest models were trained and evaluated.

After comparing the evaluation metrics, Random Forest Classifier was selected as the final model.

The Random Forest model was selected because:

- It provided better prediction capability.
 - It handled complex customer patterns effectively.
 - It provided information about important features affecting churn.
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13. Feature Importance Analysis

Feature importance analysis was performed using the Random Forest model.

This analysis helps identify the factors that have a significant impact on customer churn.

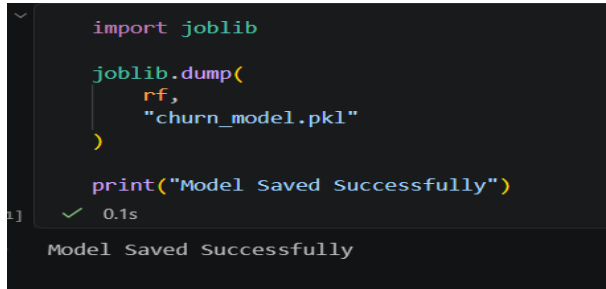
Important features identified include:

- Contract type
- Customer tenure
- Monthly charges
- Internet service
- Payment method

	Feature	Importance
19	TotalCharges	0.152603
18	MonthlyCharges	0.149936
5	tenure	0.136431
0	customerID	0.126720
15	Contract	0.085540
9	OnlineSecurity	0.044376
17	PaymentMethod	0.043136
12	TechSupport	0.036093
8	InternetService	0.023598
10	OnlineBackup	0.023139

14. Model Saving

After successful training and evaluation, the final Random Forest model was saved using Joblib.



```
import joblib

joblib.dump(
    rf,
    "churn_model.pkl"
)

print("Model Saved Successfully")
```

✓ 0.1s
Model Saved Successfully

This file can be reused for future predictions without retraining the model.

15. Results and Discussion

The project successfully developed a machine learning classification system for predicting customer churn.

The implemented models were able to analyze customer information and identify patterns associated with customer retention.

Random Forest showed better performance compared to Logistic Regression due to its ability to capture complex relationships within the dataset.

The results demonstrate that machine learning can effectively support customer retention strategies.

16. Challenges Faced

During the project development, some challenges were encountered:

- Cleaning and preparing raw customer data.
- Handling categorical variables.
- Understanding evaluation metrics.
- Selecting the most suitable algorithm.
- Comparing multiple models.

These challenges improved practical understanding of machine learning workflows.

17. Learning Outcomes

Through this project, the following concepts were learned:

- Data preprocessing techniques.
- Exploratory data analysis.
- Machine learning classification algorithms.
- Model evaluation methods.
- Feature importance interpretation.
- Saving trained machine learning models.

This project provided hands-on experience in building an end-to-end machine learning solution.

18. Future Improvements

The project can be enhanced by:

- Applying hyperparameter tuning.
 - Using advanced classification algorithms.
 - Implementing cross-validation techniques.
 - Deploying the model using Flask or FastAPI.
 - Creating a user interface for predictions.
 - Performing fairness and bias analysis.
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19. Conclusion

The Customer Churn Prediction project successfully demonstrates the application of Artificial Intelligence and Machine Learning techniques in solving real-world business problems.

The developed classification models analyze telecom customer information and predict whether customers are likely to leave the service.

Among the implemented algorithms, Random Forest was selected as the final model because of its effective performance and ability to identify complex customer patterns.

This project highlights how machine learning can help organizations make data-driven decisions and improve customer retention strategies.
